

EDA ASSIGNMENT

PROBLEM STATEMENT

- The loan-providing companies find it hard to give loans to people due to their insufficient or non-existent credit history. Because of that, some consumers use it to their advantage by becoming defaulters. Suppose you work for a consumer finance company which specializes in lending various types of loans to urban customers. It would be best if you used EDA to analyze the patterns present in the data. This will ensure that the applicants capable of repaying the loan are accepted.

BUSINESS OBJECTIVE:

➤ It aims to identify patterns which indicate if a client has difficulty paying their installments which may be used for taking actions such as denying the loan, reducing the amount of loan, lending (to risky applicants) at a higher interest rate, etc. This will ensure that the consumers capable of repaying the loan are not rejected. Identification of such applicants' using EDA is the aim of this case study. In other words, the company wants to understand the driving factors (or driver variables) behind loan default, i.e., the variables which are strong indicators of default. The company can utilize this knowledge for its portfolio and risk assessment.

TASK:

Identify Missing Data and Deal with it Appropriately: Identify the missing data in the dataset and decide on an appropriate method to deal with it.

Identify Outliers in the Dataset: Detect and identify outliers in the dataset

Analyze Data Imbalance: Determine if there is a data imbalance in the loan application dataset and calculate the ratio of data imbalance

Perform Univariate, Segmented Univariate, and Bivariate Analysis: Perform univariate analysis to understand the distribution of individual variables, segmented univariate analysis to compare variable distributions for different scenarios, and bivariate analysis to explore relationships between variables and the target variable

APPROACH:

1. Data Acquisition and Familiarization: The project began by downloading the datasets from the source. Subsequently, a thorough understanding of the datasets was gained to grasp the structure, variables, and underlying information.
2. Data Cleaning With Python: Python programming language was utilized to clean and preprocess the datasets. Tasks included handling missing data.
3. Data Analysis in Python: After completing the data cleaning process, preliminary data analysis was conducted in Python. Exploratory data analysis (EDA) techniques were applied to extract initial insights and identify any noteworthy patterns or trends.

INSIGHTS:

OUTLIERS IN THE DATASET:

- Income total has outliers indicating a presence of individuals with significantly higher incomes compared to the majority. These outliers may represent high-income professionals or individuals with substantial investments. Understanding this distribution can help tailor financial products and assess borrower risk.
- Days employed show outliers, suggesting some individuals have been employed for an unusually long time. This may indicate stable employment and a lower risk of job loss. Recognizing these outliers informs risk assessment and enables tailored loan products for long-tenured employees.
- The count of children has outliers with 19-member families, while the majority have 2 or 3 children. These outliers may reflect unique family structures or cultural dynamics. Understanding these outliers helps design loan products that cater to the specific needs of larger families.
- Count of family members exhibits outliers with 20-member families, while the majority have 3 or 4 members. These outliers may represent multigenerational households or joint families. Recognizing these outliers allows for tailored loan offerings that consider the financial dynamics of larger households.

DATA IMBALANCE:

The dataset shows a significant class imbalance, with a ratio of approximately 1:11 between non-defaulters and defaulters. Such a skewed distribution indicates that the number of default clients is relatively small compared to non-default clients.

PERFORM UNIVARIATE, SEGMENTED
UNIVARIATE, AND BIVARIATE
ANALYSIS:

When considering the clients' education type, clients with secondary education are the ones who take loans most frequently, followed by clients with higher education. Clients with secondary/secondary special and lower secondary education are the ones who face payment difficulties.

When considering the clients' credit amount, most clients with payment difficulties have loans ranging from 45,000 to 2,000,000.

When considering the annuity amount, clients with payment difficulties mostly have annuity amounts between 2,000 and 100,000.

When considering the goods price, clients who have payment difficulties tend to purchase goods within the range of 45,000 to 2,000,000.

When considering the days taken for decision, whether it is for loan approval or other purposes, the minimum and maximum time observed were 2 days and 2920 days, respectively.

CONCLUSION:

In conclusion, the insights gathered from this project shed light on various aspects of loan applicants' characteristics and behaviors. The analysis of income totals revealed the presence of outliers, indicating individuals with significantly higher incomes compared to the majority. This highlights the importance of considering income disparity when assessing borrower risk and tailoring financial products to cater to different income levels.

THANKYOU !